Question 1:

A. In a RSA cryptosystem, a participant A uses two prime numbers p = 13 and q = 17 to generate her public and private keys. If the public key of A is 35, then the private key of A is _______.

① $n = P * q = 13 * 17 = 221$

② $Q = (P-1)(q-1) = 12 * 16 = 192$

③ $e = 35$ $\quad GCD(Q, e) = 1$

$GCD(192, 35)$
$= GCD(35, 192 \bmod 35)$
$= GCD(17, 35 \bmod 17)$
$= GCD(1, 17 \bmod 1)$
$= GCD(1, 0)$

$d = (1 + k * Q)/e$

at $k=1$ $\quad d = (1+192)/35 = 5.5$ ×

at $k=2$ $\quad d = (1+192*2)/35 = 11$

d (Private key) = 11

B. Alice wants to encrypt a message to Bob by using the RSA algorithm and using keys in (A)
The plaintext = "HI".

• A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
• 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

Answer: _______________

$C(H) = 7^{35} \bmod 221$

$(7^{10} \bmod 221 * 7^{10} \bmod 221 * 7^{10} \bmod 221 * 7^{5} \bmod 221) \bmod 221 = 54$

$C(I) = 8^{35} \bmod 221$

$(8^{10} \bmod 221 * 8^{10} \bmod 221 * 8^{10} \bmod 221 * 8^{5} \bmod 221) \bmod 221 = 70$

Question 2:

Let's assume that Alice wants to establish a shared secret with Bob.
1. Alice and Bob agree on a prime number, p, and a base, g, in advance.
   For our example, let's assume that p=23 and g=5.
2. Alice chooses a secret integer a whose value is 6.
3. Bob chooses a secret integer b whose value is 15.

Find the shared key K??

Alice	BoB
$R_1 = g^a \bmod P$	$R_2 = g^b \bmod P$
$= 5^6 \bmod 23 = 8$	$= 5^{15} \bmod 23 = 19$
$\boxed{R_2 = 19}$	$\boxed{R_1 = 8}$
$K = (R_2)^a \bmod P$	$K = (R_1)^b \bmod P$
$19^6 \bmod 23 = 2$	$8^{15} \bmod 23 = 2$

Question 3:

a) Which of the following can be used for both encryption and digital signatures?

A. 3DES
B. AES
C. RSA
D. MD5

b) Digital Signatures provide which of the following?

A. Confidentiality
B. Authorization
C. Integrity
D. Authentication
E. Availability

c) The difference between a hash and a digital signature is:

A. Hashes are used to verify message authenticity, digital signatures are used to verify message integrity.
B. Hashes are used to verify message integrity and authenticity, digital signatures are used to verify message authenticity only.
C. Hashes are used to verify the message integrity only, digital signatures are used to verify message authenticity and message integrity.
D. There is no functional difference.

Question 4:

1) Which asymmetric algorithm is used for establishing a shared key?

A. Elliptic Curve
B. PGP
C. Diffie-Hellman
D. RSA

2) What is used to create a message digest that will be attached to data at rest or in transit?

A. Asymmetric algorithm
B. Symmetric algorithm

C. Encryption
D. Hashing
E. Non-repudiation
F. Authentication

3) Which is the most secure method of exchanging secret keys?

A. Text
B. Email
C. Phone
D. Diplomatic Bags
E. Asymmetric Algorithm

4) Which of these would be the PRIMARY reason we would choose to use hash functions?

A. Authorization
B. Integrity
C. Confidentiality
D. Availability

5) We have decided to change the type of hashing we use to a newer version that is collision resistant. What happens when a hash collision occurs?

A. You can figure out the plain text from the hash.
B. When two different plaintexts produce the same hash.
C. The same plain text produces two different hashes using the same hash function.
D. A variable-length text produces a fixed-length hash.

6) "Alice" and "Bob" are talking about hashing and they use the abbreviation MAC. What are they talking about?

A. Message Authentication Code.
B. Media Access Control.
C. Mandatory Access Control.
D. Message Access Code.

7) Which one of the following is not one of the basic requirements for a cryptographic hash function?
A. The function must be collision free.
B. The function must work on fixed-length input.
C. The function must be relatively easy to compute for any input.
D. The function must be one way.

8) Bob wants to produce a message digest of a 2,048-byte message he plans to send to Alice. If he uses the SHA-1 hashing algorithm, what size will the message digest for this particular message is?
A. 2048 bits
B. 1024 bits
C. 160 bits
D. 512 bits

9) What kind of attack makes the Caesar cipher virtually unusable?

A. Escrow attack
B. Man-in-the-middle attack
C. Transposition attack
D. Frequency Analysis

10) When an attacker is using a brute force attack to break a password, what are they doing?

A. Trying to recover the key without breaking the encryption.
B. Looking at the hash values and comparing it to thousands or millions of pre-calculated hashes.
C. Looking at common letter frequency to guess the plaintext.
D. Trying every possible key to, over time, break any encryption

11) If a 2,048-bit plaintext message were encrypted with the El Gamal public key cryptosystem, how long would the resulting ciphertext message be?

A. 2048 bits
B. 1024 bits
C. 4096 bits
D. 8192 bits

12) We have 100 users all need to communicate with each other. If we are using asymmetric encryption how many keys would we need?

A. 4950
B. 300
C. 2000
D. 200

13) When we are using frequency analysis, what are we looking at?
A. How often certain letters are used.
B. How often messages are sent.
C. How often pairs of letters are used.
D. How many messages are sent.